$$= \quad 200+8x$$

But, the area of the trapezium $= 352\ sq.\ cm$ (given)

Therefore, $200 + 8x = 352$

$$\Rightarrow 8x = 352 - 200$$

$$\Rightarrow 8x = 152$$

$$\Rightarrow x = \frac{152}{8}$$

$$\Rightarrow x = 19$$

Therefore, the length of the other side is 19 cm.

Think

1. Can you find the perimeter of the trapezium? Discuss.
2. Mention any three life situations where the isosceles trapeziums are used.

Example 2.14

The collar of a shirt is in the form of isosceles trapezium whose parallel sides are 17 cm and 14 cm and the distance between them is 4 cm. Find the area of canvas that will be used to stitch the collar.

Solution

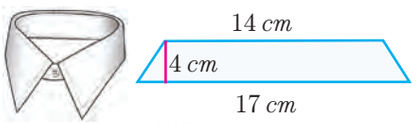


Fig.2.27

Given height $(h) = 4\ cm$

Parallel sides are $(a) = 17\ cm$ and $(b) = 14\ cm$

$$\text{Area of the trapezium} = \frac{1}{2} \times h\ (a+b)\ sq.\ units$$

$$= \frac{1}{2} \times 4\ (17+14)$$

$$= \frac{1}{2} \times 4\ (31)$$

$$= 62\ sq.\ cm$$

Therefore, the area of canvas used is 62 $sq.\ cm$.



DO YOU KNOW?

Though there are various common types of cross sections available for irrigation canals, trapezoidal cross section is widely used, because it prevents overflowing during heavy rains, maximum water flow is possible with minimum time (least frictional resistance) and safety measures can be taken easily while someone or something falls into it.

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